

Text File Conversion to Hex Code and Data Recovery

IT110 Lab Exercise

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Lab Objectives

In this laboratory exercise, you will learn to:

- Parse text and image files in Hexadecimal.
- Understand how files look in their machine-native representation.
- Apply basic Linux shell commands.

Real-world Applications:

- IT Forensics
- Data Restoration
- Virus/Malware Detection

Part I: Basic Shell Commands

Step 0: Navigation

Open your terminal. You will see a blinking cursor waiting for input. Navigate to the directory containing your file.

Type the following:

```
cd ~/lab_exercise/text_file/
```

Q: What changed in the terminal?

Part I: Basic Shell Commands

Step 0: Navigation

Open your terminal. You will see a blinking cursor waiting for input. Navigate to the directory containing your file.

Type the following:

```
cd ~/lab_exercise/text_file/
```

Q: What changed in the terminal?

Answer

The path displayed before your cursor (the prompt) should update to show the new directory.

Step 1: Listing Directory Contents

See what is inside the folder by typing:

```
ls -alth
```

Q: What do you see?

Step 1: Listing Directory Contents

See what is inside the folder by typing:

```
ls -alth
```

Q: What do you see?

- You should see a list of files with permissions, dates, and sizes.
- Look for: `helloworld.txt`

Part II: Identifying File Types

Step 0: Identify format

Verify the file type using the command line:

```
file helloworld.txt
```

Q: What was the output?

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Step 0: Identify format

Verify the file type using the command line:

```
file helloworld.txt
```

Q: What was the output?

- It should identify the file as ASCII text.

Dumping to Hexadecimal

Step 1: View Hex Dump

Type the following to dump the text file into hexadecimal encoding:

```
xxd -C helloworld.txt
```

Q: What was the output?

- You see the hexadecimal values on the left and the ASCII representation on the right.

Saving the Hex Output

Step 2: Save to File

Save the "plain" hex dump ('-p') into a new text file using the redirection operator ('>'):

```
xxd -p helloworld.txt > hexdump_TXT.txt
```

Command Breakdown:

- `xxd`: The conversion tool.
- `-p`: "Plain" output (continuous hex string).
- `>`: Redirects output from the screen to a file.

Verifying the Output

Now, verify the content of your new file:

```
cat hexdump_TXT.txt
```

Q: What happens when you type that?

- The cat command parses the text file and displays the raw ASCII content (the hex string) in the terminal.

Try opening this file in the GUI text editor to see if it matches!

Part III: Reverting Hex to Text

Step 0: Revert to Binary

Convert the hexadecimal text file back into its original text format:

```
xxd -r -p hexdump_TXT.txt > output.txt
```

Command Breakdown:

- `-r`: Revert (Reverse operation).
- `-p`: Read plain hex format.

Part III: Reverting Hex to Text

Step 0: Revert to Binary

Convert the hexadecimal text file back into its original text format:

```
xxd -r -p hexdump_TXT.txt > output.txt
```

Command Breakdown:

- `-r`: Revert (Reverse operation).
- `-p`: Read plain hex format.

Step 1: Verify

```
cat output.txt
```

Q: What was the output this time?

It should be the original "Hello World" text.

Section B: Image Files

Step 0: Navigation

Move to the directory containing images:

```
cd ..  
cd text_file % Note: Ensure this is the correct  
              folder name  
ls -alth
```

Step 1: Inspect Data

```
file 16x16_16ppi_no_compression_gryascale.jpeg  
file 16x16_16ppi_no_compression_grayscale.tiff
```

- **TIFF:** Uncompressed. Every pixel is a hex integer.
- **JPEG:** Lossy compression (Discrete Cosine Transform).

Analyzing JPEG Headers

Step 2: JPEG Hex Dump

Type the following:

```
hexdump -vC 16x16_16ppi_no_compression_gryascale.jpeg
```

Flags:

- -v: Verbose (show everything).
- -C: Canonical hex+ASCII display.

Step 3: Magic Numbers Look at the very beginning of the output. **FFD8** is the "Signature" that tells the OS "This is a JPEG".

Analyzing TIFF Patterns

Step 4: Dump TIFF

```
xxd -p 16x16_16ppi_no_compression_grayscale.tiff >  
hexdump_TIFF.txt
```

Step 5: Inspect Pattern Open `hexdump_TIFF.txt` in a text editor.

Scroll until you see a pattern like:

00 00 00 00... FF FF...

Q: What do you think that is?

Analyzing TIFF Patterns

Step 4: Dump TIFF

```
xxd -p 16x16_16ppi_no_compression_grayscale.tiff >  
    hexdump_TIFF.txt
```

Step 5: Inspect Pattern Open `hexdump_TIFF.txt` in a text editor.

Scroll until you see a pattern like:

```
00 00 00 00... FF FF...
```

Q: What do you think that is?

Answer

Since TIFF is uncompressed, you are looking at the raw pixel data (black vs. white pixels).

File Terminators & Reverting

Step 6: Identify Terminator Run hexdump on the JPEG again. Look for the end of the file.

The binary string **FFD9** indicates the end of a JPEG file.

Step 7: Convert Hex to Image (*Ensure you have a hexdump_JPEG.txt file first*)

```
xxd -r -p hexdump_JPEG.txt output.jpeg
```

Q: Is there pixel information (00 00 FF FF) visible in the JPEG hex?
No. Because JPEG uses compression, the raw pixel map is destroyed and encoded mathematically.

Appendix A: Windows Alternatives

If you are using Windows PowerShell (without WSL), use these commands:

1. Listing Files (ls)

```
dir
```

2. Reading Files (cat)

```
type filename.txt
```

3. Hex Dump (xxd) Windows does not have xxd by default. Use PowerShell 5.0+:

```
Format-Hex helloworld.txt
```

4. Creating a Hex Dump File

```
Format-Hex helloworld.txt | Select-Object -  
ExpandProperty Bytes | ForEach-Object { "{0:X2}" -f  
$_ } | Out-File hexdump.txt -NoNewline
```

Appendix B: Google Colab Alternatives

Google Colab runs on Linux, so you can use the exact same commands by adding an exclamation mark (!) before them.

1. List files

```
!ls -alth
```

2. Dump to Hex

```
!xxd -p helloworld.txt > hexdump_TXT.txt
```

3. Revert to Text

```
!xxd -r -p hexdump_TXT.txt output.txt
```

4. Verify

```
!cat output.txt
```

Questions?